Exploratory Data Analysis (EDA) on Airbnb NYC Dataset

Introduction

Exploratory Data Analysis (EDA) helps us understand the structure, patterns, and insights from a dataset.

In this project, we analyze the Airbnb NYC dataset to explore pricing patterns, room types, neighborhood distribution, and availability of listings.

Importing Required Libraries

We first import the libraries needed for data analysis and visualization.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Loading the Dataset

The dataset is loaded using pandas and the first few rows are displayed.

```
df = pd.read_csv("AB_NYC_2019.csv.zip")
df.head()
```

```
     id                                                name  host_id  \
0  2539                Clean & quiet apt home by the park     2787
1  2595                             Skylit Midtown Castle     2845
2  3647               THE VILLAGE OF HARLEM....NEW YORK !     4632
3  3831                   Cozy Entire Floor of Brownstone     4869
4  5022  Entire Apt: Spacious Studio/Loft by central park     7192

     host_name neighbourhood_group neighbourhood  latitude  longitude
\
0         John            Brooklyn    Kensington  40.64749  -73.97237

1     Jennifer           Manhattan       Midtown  40.75362  -73.98377

2    Elisabeth           Manhattan        Harlem  40.80902  -73.94190

3  LisaRoxanne            Brooklyn  Clinton Hill  40.68514  -73.95976

4        Laura           Manhattan   East Harlem  40.79851  -73.94399
```

```
         room_type  price  minimum_nights  number_of_reviews
last_review  \
0     Private room    149               1                  9  2018-10-
19
1  Entire home/apt    225               1                 45  2019-05-
21
2     Private room    150               3                  0
NaN
3  Entire home/apt     89               1                270  2019-07-
05
4  Entire home/apt     80              10                  9  2018-11-
19

   reviews_per_month  calculated_host_listings_count  availability_365

0               0.21                               6               365

1               0.38                               2               355

2                NaN                               1               365

3               4.64                               1               194

4               0.10                               1                 0
```

Dataset Overview

We examine the number of rows, columns, and data types in the dataset.

```
df.shape
```

```
(48895, 16)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 48895 entries, 0 to 48894
Data columns (total 16 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   id                              48895 non-null  int64
 1   name                            48879 non-null  object
 2   host_id                         48895 non-null  int64
 3   host_name                       48874 non-null  object
 4   neighbourhood_group             48895 non-null  object
 5   neighbourhood                   48895 non-null  object
 6   latitude                        48895 non-null  float64
```

```
 7   longitude                       48895 non-null  float64
 8   room_type                       48895 non-null  object
 9   price                           48895 non-null  int64
 10  minimum_nights                  48895 non-null  int64
 11  number_of_reviews               48895 non-null  int64
 12  last_review                     38843 non-null  object
 13  reviews_per_month               38843 non-null  float64
 14  calculated_host_listings_count  48895 non-null  int64
 15  availability_365                48895 non-null  int64
dtypes: float64(3), int64(7), object(6)
memory usage: 6.0+ MB
```

Checking Missing Values

Missing values can affect analysis results, so we identify them first.

```
df.isnull().sum()
```

```
id                                    0
name                                 16
host_id                               0
host_name                            21
neighbourhood_group                   0
neighbourhood                         0
latitude                              0
longitude                             0
room_type                             0
price                                 0
minimum_nights                        0
number_of_reviews                     0
last_review                       10052
reviews_per_month                 10052
calculated_host_listings_count        0
availability_365                      0
dtype: int64
```

Data Cleaning

We handle missing values to ensure accurate analysis.

```
df['reviews_per_month'] = df['reviews_per_month'].fillna(0)
df['last_review'] = df['last_review'].fillna("No Review")
df['name'] = df['name'].fillna("Unknown")
df['host_name'] = df['host_name'].fillna("Unknown")
```

Descriptive Statistics

We analyze basic statistical properties such as mean, median, minimum, and maximum values.

```
df.describe()
```

```
                  id       host_id      latitude     longitude
price  \
count  4.889500e+04  4.889500e+04  48895.000000  48895.000000
48895.000000
mean   1.901714e+07  6.762001e+07     40.728949    -73.952170
152.720687
std    1.098311e+07  7.861097e+07      0.054530      0.046157
240.154170
min    2.539000e+03  2.438000e+03     40.499790    -74.244420
0.000000
25%    9.471945e+06  7.822033e+06     40.690100    -73.983070
69.000000
50%    1.967728e+07  3.079382e+07     40.723070    -73.955680
106.000000
75%    2.915218e+07  1.074344e+08     40.763115    -73.936275
175.000000
max    3.648724e+07  2.743213e+08     40.913060    -73.712990
10000.000000

       minimum_nights  number_of_reviews  reviews_per_month  \
count    48895.000000       48895.000000       48895.000000
mean         7.029962          23.274466           1.090910
std         20.510550          44.550582           1.597283
min          1.000000           0.000000           0.000000
25%          1.000000           1.000000           0.040000
50%          3.000000           5.000000           0.370000
75%          5.000000          24.000000           1.580000
max       1250.000000         629.000000          58.500000

       calculated_host_listings_count  availability_365
count                    48895.000000      48895.000000
mean                         7.143982        112.781327
std                         32.952519        131.622289
min                          1.000000          0.000000
25%                          1.000000          0.000000
50%                          1.000000         45.000000
75%                          2.000000        227.000000
max                        327.000000        365.000000
```

Distribution of Room Types

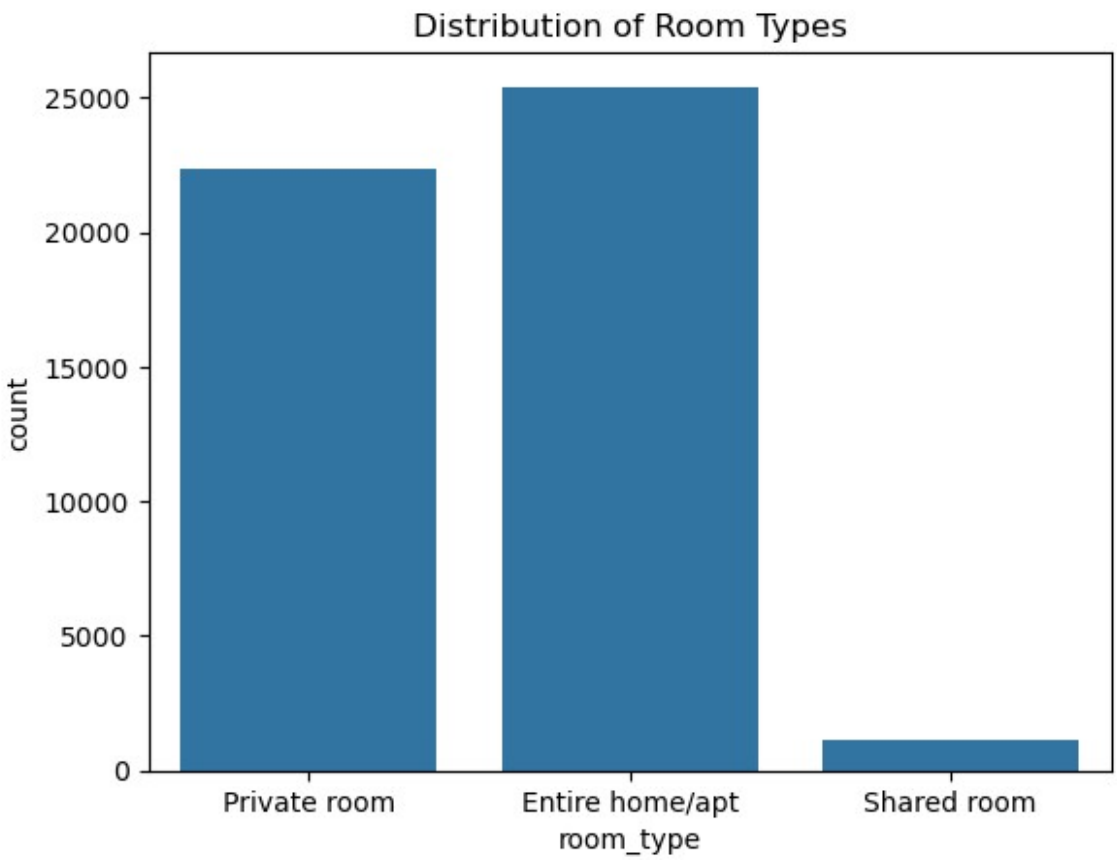
We analyze how many listings belong to each room type.

```
df["room_type"].value_counts()
```

```
room_type
Entire home/apt    25409
```

```
Private room       22326
Shared room         1160
Name: count, dtype: int64
```

```
sns.countplot(x="room_type", data=df)
plt.title("Distribution of Room Types")
plt.show()
```

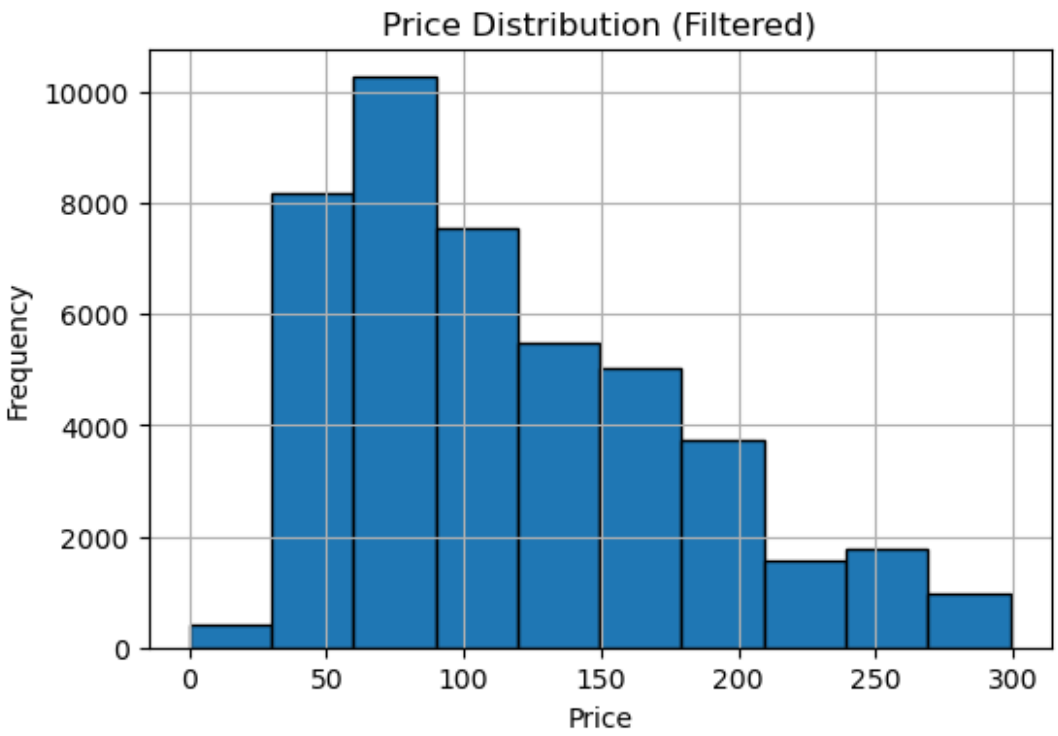


Observation: This chart shows which type of accommodation (entire home, private room, shared room) is most common.

Price Distribution

We examine how listing prices are distributed.

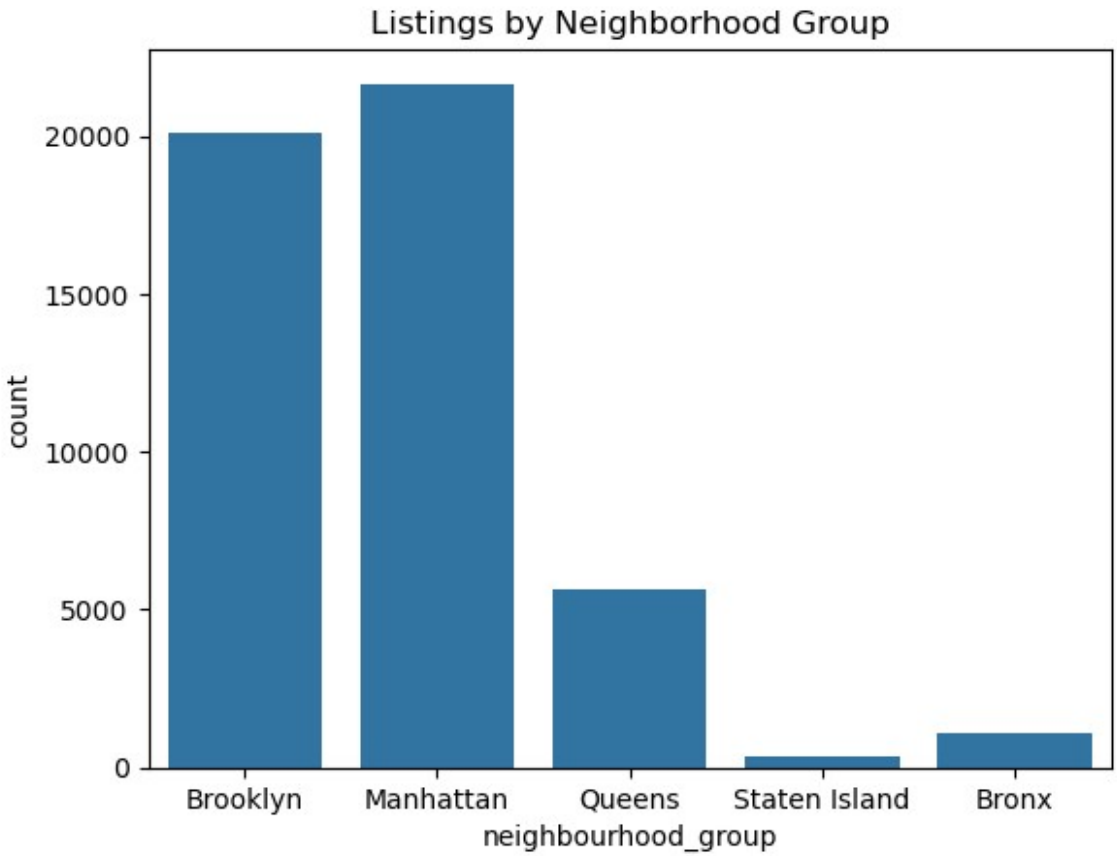
```
df_filtered = df[df["price"] <300]

plt.figure(figsize=(6,4))
df_filtered["price"].hist(edgecolor='black')
plt.title("Price Distribution (Filtered)")
plt.xlabel("Price")
plt.ylabel("Frequency")
plt.show()
```



Listings by Neighborhood Group

We analyze which borough of New York has the most Airbnb listings.

```
sns.countplot(x="neighbourhood_group", data=df)
plt.title("Listings by Neighborhood Group")
plt.show()
```

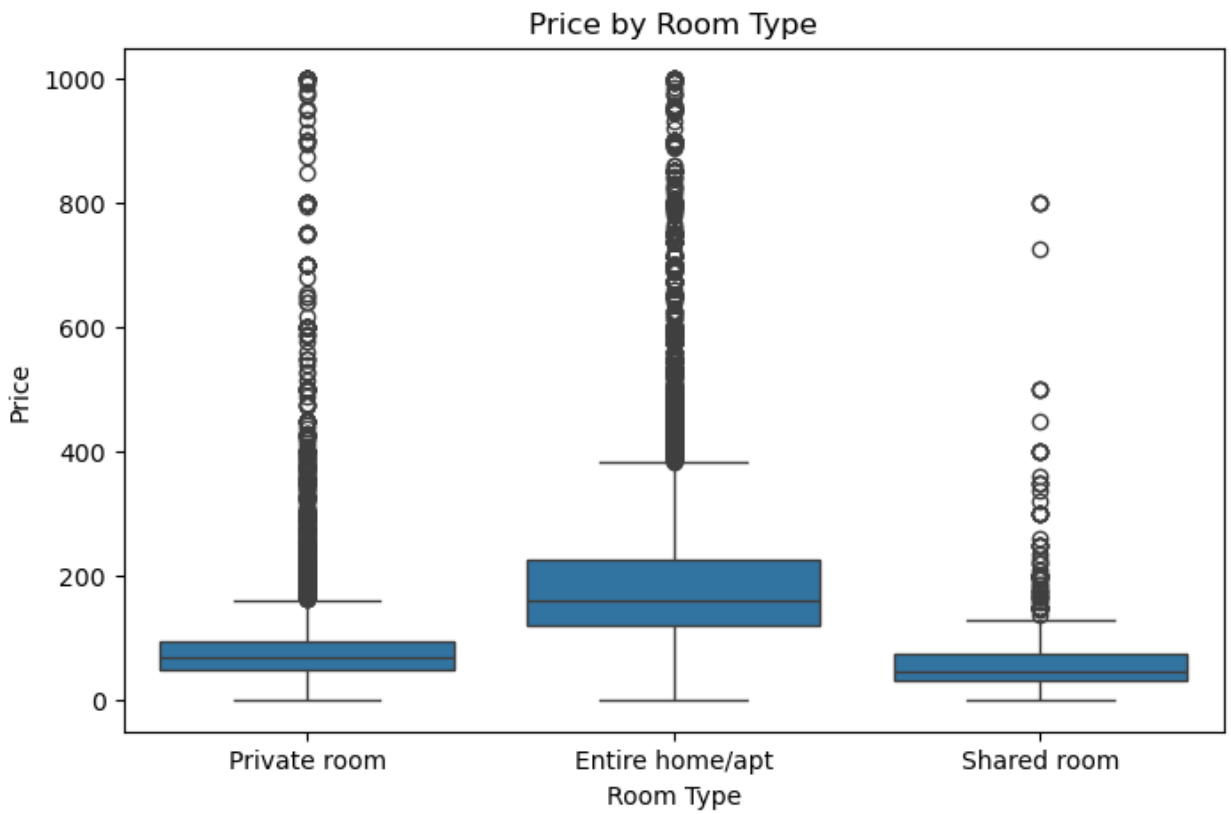


Observation: This graph shows which borough has the highest number of Airbnb listings.

Price by Room Type

We compare prices across different types of rooms.

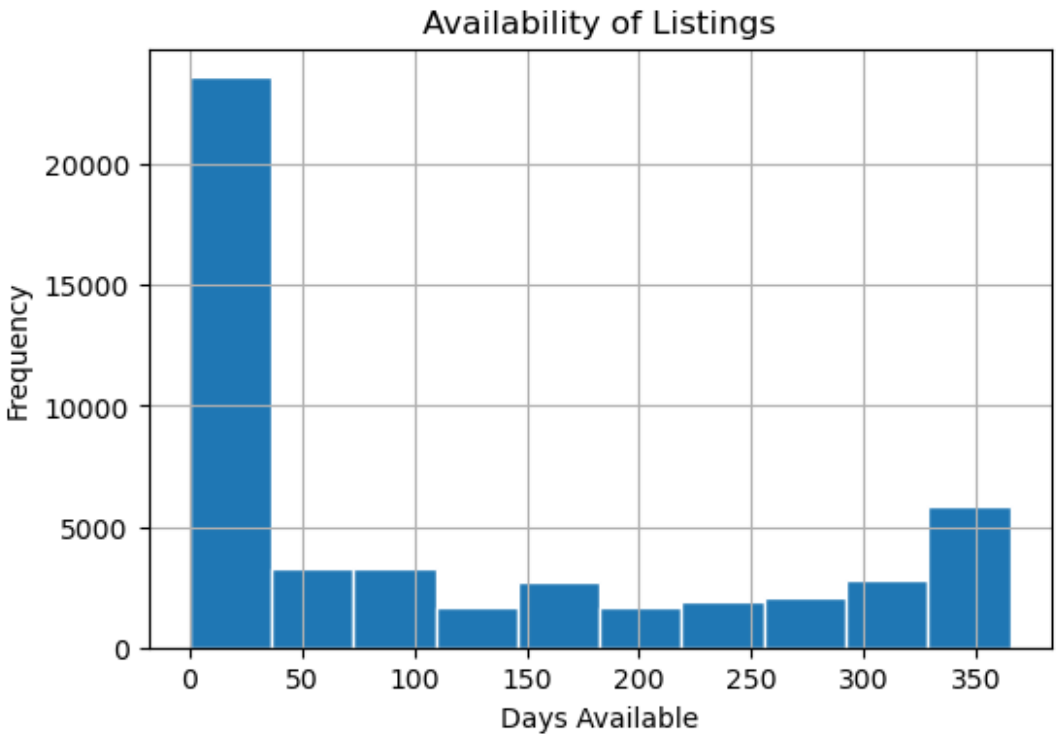
```
filtered_df = df[df["price"] < 1000]



plt.figure(figsize=(8,5))
sns.boxplot(x="room_type", y="price", data=filtered_df)

plt.title("Price by Room Type")
plt.xlabel("Room Type")
plt.ylabel("Price")
plt.show()
```



Availability Analysis

We analyze how many days listings are available throughout the year.

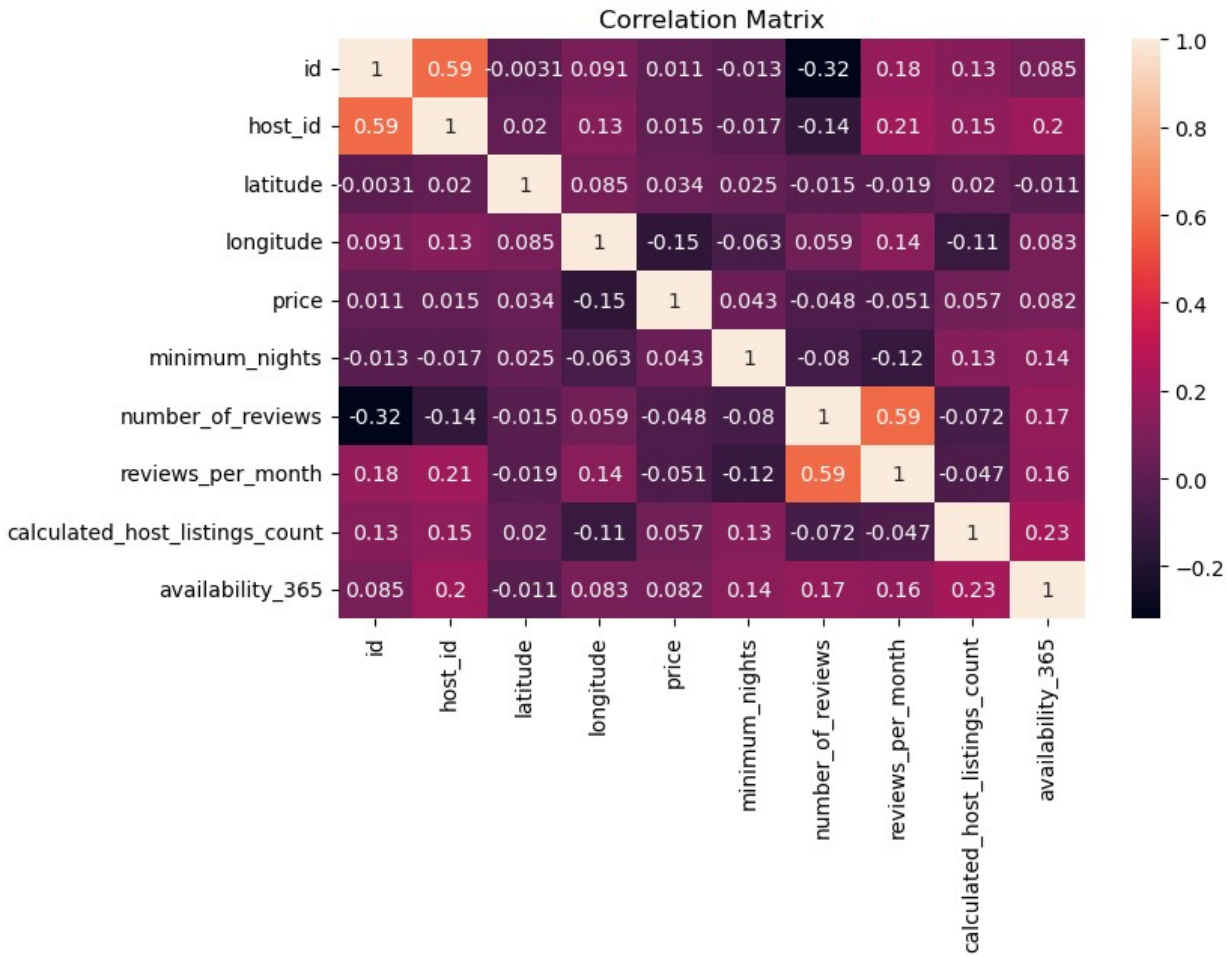
```
plt.figure(figsize=(6,4))
df["availability_365"].hist(edgecolor='white')
plt.title("Availability of Listings")
plt.xlabel("Days Available")
plt.ylabel("Frequency")
plt.show()
```



Correlation Analysis

A heatmap helps identify relationships between numerical variables.

```
plt.figure(figsize=(8,5))
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Matrix")
plt.show()
```



Key Insights

1. Entire homes/apartments are the most common Airbnb listings.
2. Manhattan have significantly more listings than others.
3. Prices vary widely depending on room type and location.
4. Many listings are available throughout most of the year.

Recommendations

• Property owners should price listings competitively based on room type.
• High-demand neighborhoods can be targeted for investment.
• Hosts can optimize availability to maximize bookings.